

Multi-User ATM Machine

RTL Design and Functional Verification Using Verilog HDL

A Finite State Machine (FSM) Based Banking Transaction System
for Secure Multi-User Authentication, Cash Management,
PIN Verification, and Transaction Processing

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Abstract

The **Multi-User ATM Machine** is a Finite State Machine (FSM) based banking transaction system designed and implemented using **Verilog HDL**. The project simulates the working of an Automated Teller Machine by providing secure authentication and banking operations for multiple users through a single RTL design.

The system performs card validation and PIN verification before allowing users to access banking services such as cash withdrawal, cash deposit, balance enquiry, and PIN change. Each user has an independent card number, PIN, and account balance, while the ATM maintains a common cash balance and note inventory. During withdrawal, the controller verifies both the account balance and ATM cash availability before dispensing 500, 200, and 100 notes. Deposit transactions update both the user's account balance and the ATM cash inventory.

To improve security, the ATM locks a card after three consecutive incorrect PIN attempts and restores access after a predefined lock duration. The complete design is implemented using FSM states for authentication, transaction processing, security handling, and card ejection.

The functionality of the ATM controller is verified using the **Vivado Simulation Environment** through various test cases, including successful transactions, invalid card detection, incorrect PIN attempts, cash withdrawal, cash deposit, balance enquiry, PIN change, and card lock scenarios. This project demonstrates the application of RTL design, sequential logic, and FSM-based control in developing a secure digital banking system.

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1 Introduction

An Automated Teller Machine (ATM) is an electronic banking system that enables customers to perform financial transactions such as cash withdrawal, cash deposit, balance enquiry, and PIN change without visiting a bank. The complete operation of an ATM follows a sequence of authentication and transaction states, making it a suitable application for Finite State Machine (FSM) based digital design.

This project presents a **Multi-User ATM Machine** implemented using **Verilog HDL**. The controller authenticates users through card validation and PIN verification before allowing banking operations. It supports multiple users by maintaining separate card numbers, PINs, and account balances while sharing a common ATM cash balance and currency note inventory.

The ATM controller performs withdrawal, deposit, balance enquiry, PIN modification, card ejection, and temporary card locking after three incorrect PIN attempts. Cash withdrawal is completed only after verifying both the user's account balance and the ATM's available cash balance, along with the availability of 500, 200, and 100 notes.

The design is implemented using RTL methodology and verified using the Vivado simulation environment through a Verilog testbench. This project demonstrates the application of Verilog HDL, sequential logic design, and FSM-based control in developing a secure, efficient, and reliable digital banking system.

2 Project Overview

This section presents the objectives and technical specifications of the proposed Multi-User ATM Machine, providing a high-level overview of its design and banking functionalities implemented using Verilog HDL.

2.1 Objectives

- Design a Multi-User ATM controller using Verilog HDL.
- Implement secure card authentication and PIN verification.
- Perform cash withdrawal, cash deposit, and balance enquiry operations.
- Support PIN change functionality for authenticated users.

- Maintain ATM cash balance and currency note inventory.
- Implement card lock security after three consecutive incorrect PIN attempts.
- Functionally verify the complete design using the Vivado simulation environment.

3 System Architecture

The Multi-User ATM Machine is designed using a Finite State Machine (FSM) architecture in Verilog HDL. The system consists of authentication, transaction processing, cash management, and security units that work together to perform banking operations for multiple users. The FSM controls the complete transaction flow by transitioning between different states based on user inputs and transaction conditions.

3.1 Overall Block Diagram

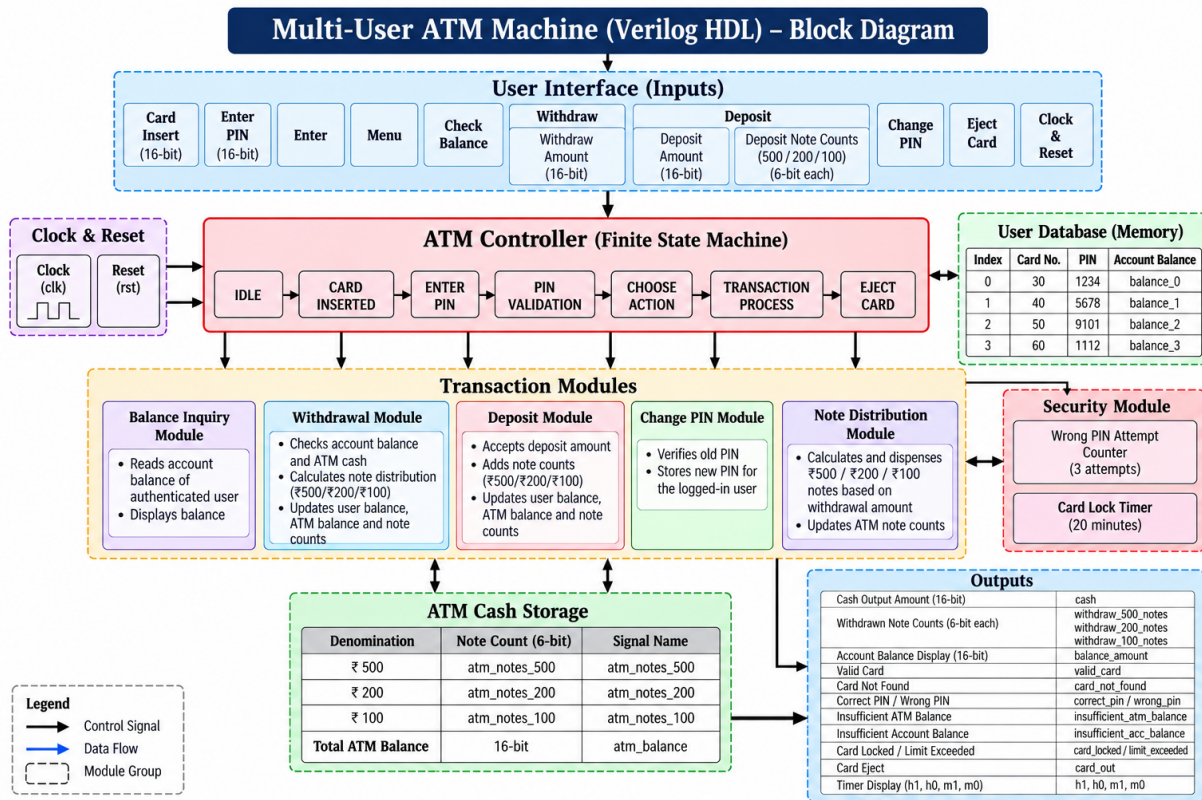


Figure 1: Overall Block Diagram of the Multi-User ATM Machine.

The block diagram illustrates the overall architecture of the ATM controller. User inputs such as card number, PIN, withdrawal amount, deposit amount, and transaction control signals are provided to the FSM-based ATM controller. The controller communicates with the user account database, ATM cash management unit, and security logic to execute banking transactions and generate the required outputs.

3.2 Finite State Machine Diagram

The Multi-User ATM Machine is implemented using an FSM with eleven states. Each state represents a specific banking operation, including authentication, transaction processing, security handling, and card ejection. The controller transitions between states according to user inputs and transaction results.

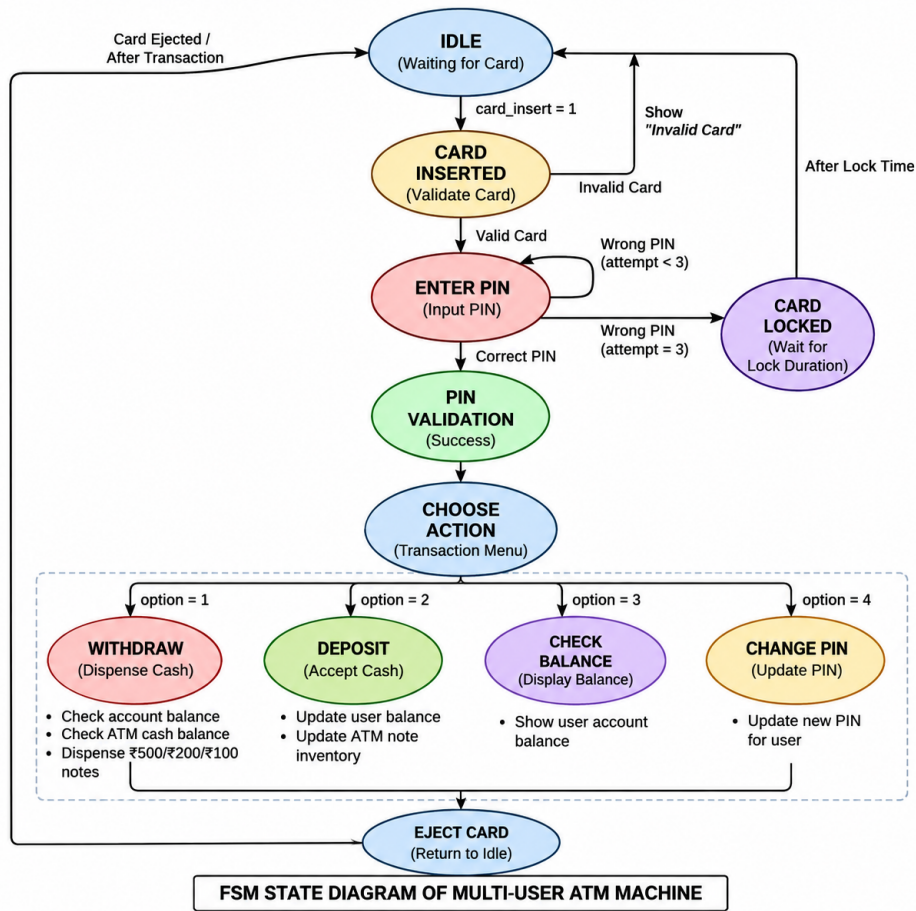


Figure 2: Finite State Machine (FSM) of the Multi-User ATM Machine.

The FSM starts from the **IDLE** state and waits for card insertion. After successful card validation and PIN authentication, the controller enters the **CHOOSE_ACTION** state, where the user can perform withdrawal, deposit, balance enquiry, or PIN change operations. Invalid authentication moves the controller

to the **CARD_LOCKED** state after three incorrect PIN attempts, while transaction completion returns the FSM to the **IDLE** state through the **EJECT_CARD** state.

3.3 System Operation

The ATM controller operates through a sequence of FSM states to perform secure banking transactions. Initially, the controller remains in the **IDLE** state and waits for card insertion. After validating the card number and verifying the user's PIN, the FSM enters the **CHOOSE_ACTION** state, where the user can select withdrawal, deposit, balance enquiry, PIN change, or card ejection.

During a withdrawal, the controller verifies both the user's account balance and the ATM cash balance before dispensing the required 500, 200, and 100 notes. Deposit transactions update the user's account balance and ATM cash inventory. If three consecutive incorrect PIN attempts are detected, the controller enters the **CARD_LOCKED** state for a predefined lock duration before returning to the **IDLE** state.

4 Design Implementation

The Multi-User ATM Machine is implemented using a Finite State Machine (FSM) in Verilog HDL. The controller is designed as a single top-level module named **atm_demo**, where all banking operations are executed through sequential FSM state transitions. Each state performs a specific operation such as authentication, transaction processing, security handling, or card ejection.

The FSM consists of eleven states: **IDLE**, **CARD_INSERTED**, **ENTER_PIN**, **PIN_VALIDATION**, **CHOOSE_ACTION**, **WITHDRAW**, **CHECK_BALANCE**, **DEPOSIT**, **CHANGE_PIN**, **CARD_LOCKED**, and **EJECT_CARD**. The controller transitions between these states based on user inputs and transaction conditions.

4.1 IDLE State

The **IDLE** state is the initial state of the ATM controller. In this state, the system waits for the user to insert an ATM card and resets the authentication signals. When a card is inserted, the FSM transitions to the **CARD_INSERTED** state.

4.2 CARD_INSERTED State

The CARD_INSERTED state validates the inserted card by comparing it with the stored card numbers of all registered users. If the card is valid, the authenticated user is selected and the FSM moves to the ENTER_PIN state; otherwise, the `card_not_found` signal is generated.

4.3 ENTER_PIN State

The ENTER_PIN state waits for the user to enter the PIN associated with the authenticated card. Once the `enter` signal is received, the FSM proceeds to the PIN_VALIDATION state for authentication.

4.4 PIN_VALIDATION State

The PIN_VALIDATION state compares the entered PIN with the stored PIN of the authenticated user. If the PIN is correct, the FSM transitions to the CHOOSE_ACTION state. Otherwise, the incorrect PIN attempt counter is updated, and after three consecutive incorrect attempts, the controller enters the CARD_LOCKED state.

4.5 CHOOSE_ACTION State

The CHOOSE_ACTION state displays the available banking operations after successful authentication. Based on the user's selection, the FSM moves to the WITHDRAW, DEPOSIT, CHECK_BALANCE, CHANGE_PIN, or EJECT_CARD state.

4.6 WITHDRAW State

The WITHDRAW state processes the cash withdrawal request after verifying the user's account balance and the ATM cash balance. If sufficient funds are available, the user's account balance, ATM cash balance, and currency note inventory are updated before returning to the CHOOSE_ACTION state. CHOOSE_ACTION state.

4.7 CHECK_BALANCE State

The CHECK_BALANCE state displays the current account balance of the authenticated user. The controller remains in this state until the user selects another transaction or ejects the card.

4.8 DEPOSIT State

The DEPOSIT state adds the deposited amount to the user's account balance and updates the ATM cash balance and currency note inventory. After the deposit transaction is completed, the FSM returns to the **CHOOSE_ACTION** state.

4.9 CHANGE_PIN State

The CHANGE_PIN state allows the authenticated user to update the existing PIN with a new PIN. After the PIN is successfully updated, the controller returns to the **CHOOSE_ACTION** state.

4.10 CARD_LOCKED State

The CARD_LOCKED state is activated after three consecutive incorrect PIN attempts. The card remains locked for a predefined lock duration of 20 minutes, after which the lock status is cleared and the controller returns to the **IDLE** state.

4.11 EJECT_CARD State

The EJECT_CARD state completes the current transaction and ejects the user's card. After the card is removed, the ATM controller returns to the **IDLE** state and becomes ready for the next user.

5 Functional Verification

Functional verification is carried out using a Verilog testbench in the Vivado Simulator. The testbench applies different banking transaction scenarios to verify the FSM state transitions, card authentication, PIN verification, cash transactions, balance updates, and security features implemented in the ATM controller.

5.1 Test Scenarios

The following test scenarios are used to verify the functionality of the Multi-User ATM Machine.

Table 1: Functional Verification Test Scenarios

Test Case	Description
TC1	Valid card insertion and successful PIN authentication.
TC2	Cash withdrawal with sufficient account balance and ATM cash balance.
TC3	Balance enquiry after a successful transaction.
TC4	Three consecutive incorrect PIN attempts leading to card lock.
TC5	Cash deposit and account balance update.
TC6	PIN change and login using the updated PIN.
TC7	Invalid card detection using an unregistered card number.
TC8	Cash withdrawal with insufficient ATM cash balance.
TC9	Cash withdrawal with insufficient account balance.

6 Results

The Multi-User ATM Machine was successfully synthesized and verified using Vivado. The RTL schematic illustrates the synthesized hardware architecture of the ATM controller, while the Vivado simulation waveform validates the functionality of the FSM during different banking transaction scenarios.

6.1 RTL Schematic

The RTL schematic represents the synthesized hardware implementation of the Multi-User ATM Machine generated using Vivado. It shows the top-level ATM controller along with its input and output ports, internal registers, FSM control logic, and transaction processing blocks.

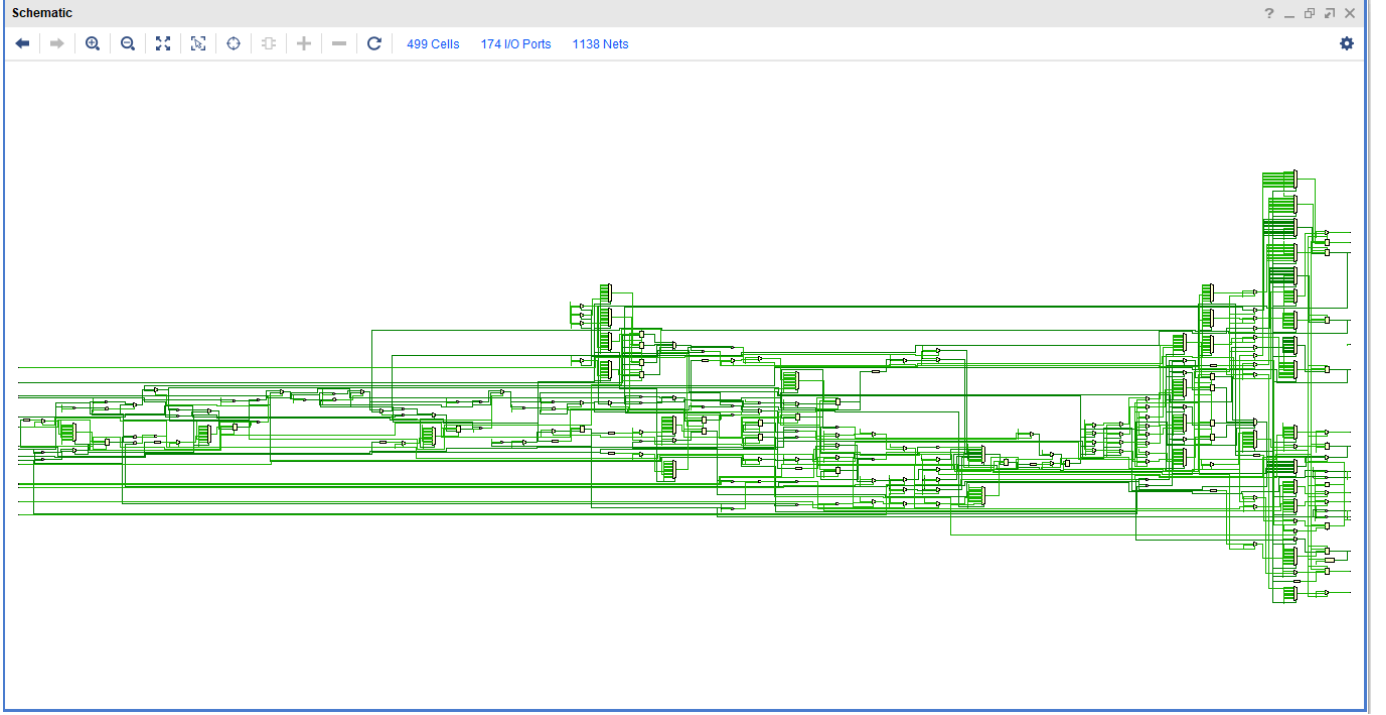


Figure 3: RTL Schematic of the Multi-User ATM Machine generated using Vivado.

The generated RTL confirms that the ATM controller is implemented as an FSM-based hardware design capable of performing authentication, transaction processing, cash management, and security operations for multiple users.

6.2 Simulation Results

The functional behavior of the Multi-User ATM Machine was verified using the Vivado Simulator. The simulation waveform demonstrates the execution of the FSM through different banking transaction scenarios, including card authentication, PIN verification, cash withdrawal, cash deposit, balance enquiry, PIN change, invalid card detection, and card locking.

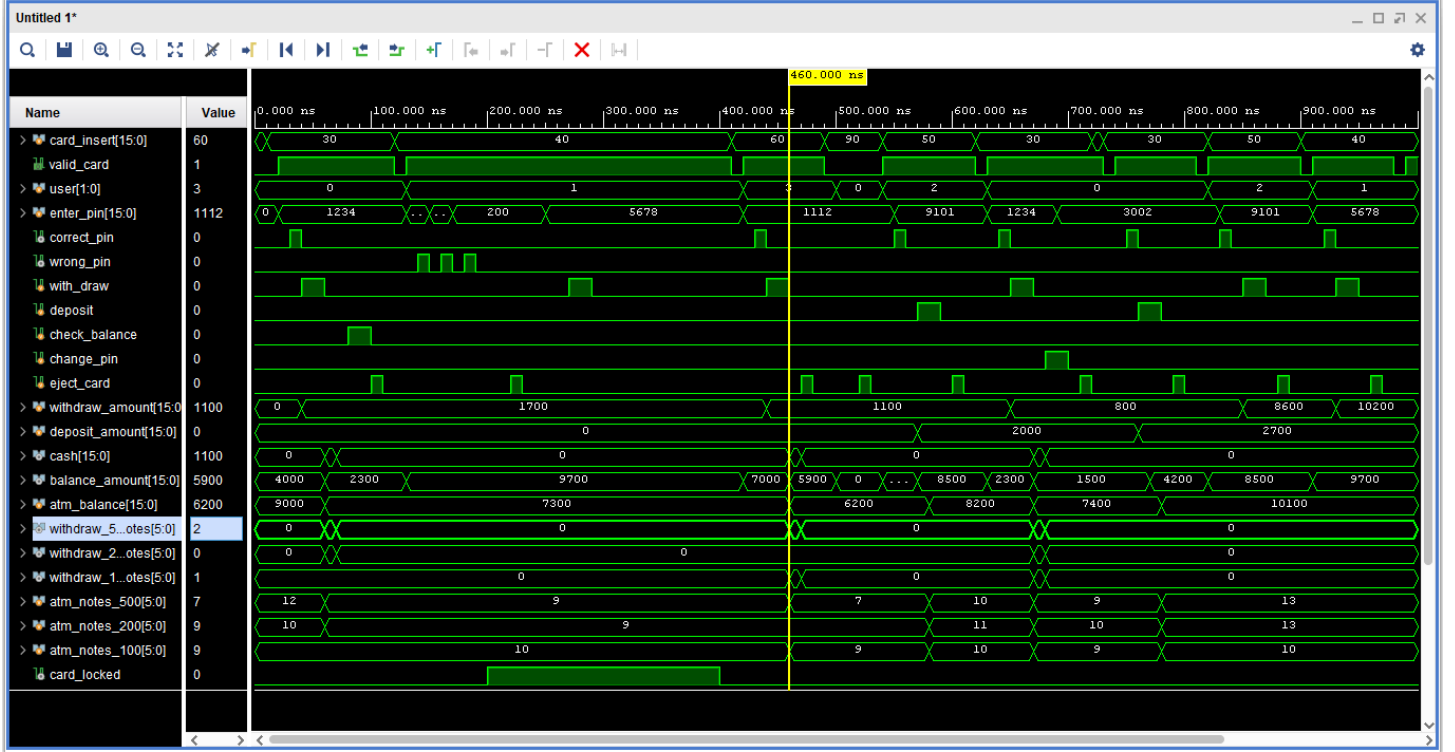


Figure 4: Vivado simulation waveform of the Multi-User ATM Machine.

The simulation waveform confirms the correct transition of the FSM between authentication, transaction processing, security handling, and card ejection states. It also verifies successful cash transactions, account balance updates, currency note distribution, and the temporary card locking mechanism after three consecutive incorrect PIN attempts.

7 Applications

The Multi-User ATM Machine can be used in various digital banking and VLSI applications. The FSM-based design provides secure authentication and reliable transaction processing for multiple users.

- ATM transaction control systems.
- Banking and financial transaction simulators.
- Digital security and authentication systems.

8 Future Enhancements

The current ATM controller can be extended with additional banking and security features in future versions.

- Support for a larger number of users using external memory.
- Daily transaction and withdrawal limit implementation.
- OTP or biometric-based user authentication.
- Mini statement and transaction history feature.
- FPGA implementation with keypad, LCD, and seven-segment display.
- Integration with a database or cloud-based banking system.

9 Conclusion

The Multi-User ATM Machine was successfully designed and implemented using Verilog HDL based on a Finite State Machine (FSM) architecture. The controller provides secure card authentication, PIN verification, cash withdrawal, cash deposit, balance enquiry, PIN change functionality, and card lock security for multiple users.

The complete design was synthesized and functionally verified using Vivado, where all transaction scenarios and FSM state transitions were successfully simulated. This project demonstrates the practical application of RTL design, sequential logic, and FSM-based control in developing a secure and efficient digital banking system.